



In March 1744, six unusual tails were seen shooting above the horizon from the amazingly bright Great Comet of 1744.

**IN SPRING 1744, NIGHT SKIES** around the world were illuminated by **one of the brightest ever comets**. It was spotted through a telescope late in 1743 by German astronomers **Jan de Munck** and **Dirk Klinkenberg**, and Swiss astronomer Jean-Philippe de Chéseaux—it later became known as the Comet de Klinkenberg-Chéseaux. By the next spring, this rare “**Great Comet**” was so bright that it outshone Venus at night and, for a few weeks in March 1744, it could even be seen by day. Thanks to new surveying equipment, it became possible to make accurate maps using



#### Cassini's map of France

In making the first detailed, accurate map of France, Cassini set out: “To measure distances by triangulation and thus establish the exact position of the settlements.”

**triangulation**—a technique that establishes positions through measuring angles. In 1744, French mapmaker **César-François Cassini de Thury** (1714–84), also known as Cassini III, began a huge project to make the **first accurate map of all France** at the scale of 1:84,600. The project was a landmark in mapmaking.

During this year, Swiss mathematician **Leonhard Euler** (1707–83) was working in Berlin on optics. The clarity of his papers helped ensure that **Huygens' theory that light travels in waves** prevailed over Isaac Newton's theory of light “corpuscles” or particles (see 1675).



French philosopher Pierre Maupertuis's ideas hinted at later evolutionary theory.

**IN 1745, SWISS NATURALIST CHARLES BONNET** (1720–93) wrote a **key study on insects**, *Traité d'Insectologie* (*Treatise on Insectology*). In it he noted that **caterpillars breathe through pores**, and that **aphids reproduce by parthogenesis** (without the need for mating).

In France that year, the mathematician and philosopher **Pierre Louis Maupertuis** (1698–1759) was writing *Vénus Physique*, in which he hinted at **ideas that emerged later in evolutionary theory**. He suggested that only those animals made in a way that best met their needs would survive, while those lacking appropriate characteristics would not. Maupertuis also suggested that **all life descends from a common ancestor**.

Also this year, French mathematician, **César-François Cassini de Thury** developed the **Cassini map projection**. All map projections are accurate but they distort in various ways. The Cassini projection was accurate at right angles to a central meridian, so was good for local grid-based maps. For this reason, the Cassini projection was used for the well-known **Ordnance Survey maps** of the UK.

In 1746, French mineralogist **Jean-Étienne Guettard** (1715–86) was pioneering

“ **THE SPECIES... TODAY ARE... SMALLEST PART OF WHAT BLIND DESTINY HAS PRODUCED...** ”

Pierre-Louis Moreau de Maupertuis, in *Essai de cosmologie*, 1750

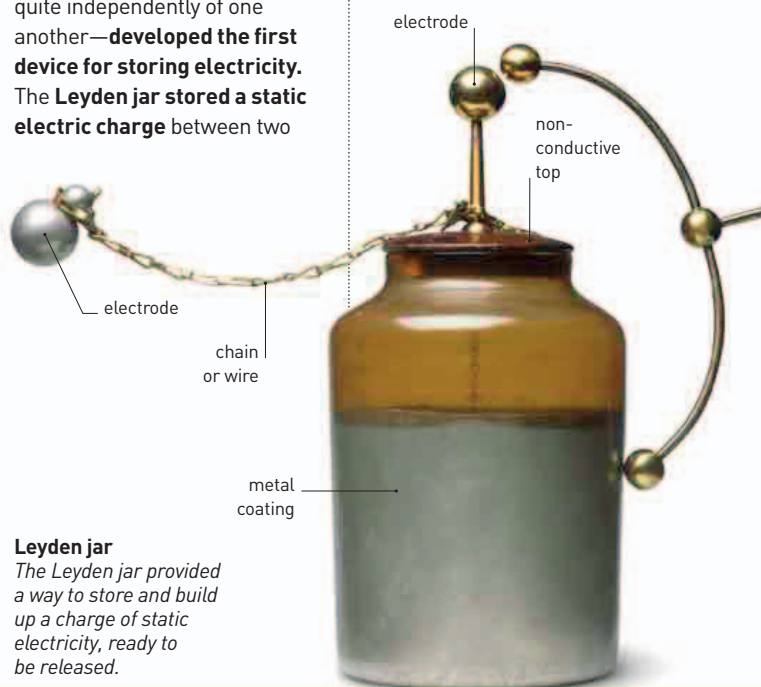
“ **I HAVE FOUND OUT SO MUCH ABOUT ELECTRICITY THAT... I UNDERSTAND NOTHING AND CAN EXPLAIN NOTHING.** ”

Pieter van Musschenbroek, Dutch physicist, 1746

another kind of map, showing all the country's surface minerals. It was perhaps the **first ever major geological map**.

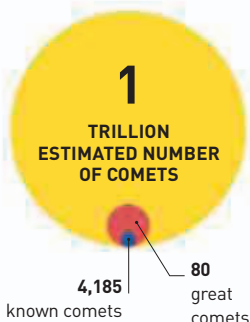
In the city of Leyden in Holland, German cleric Ewald Georg von Kleist (1700–48) and Dutch physicist Pieter van Musschenbroek (1692–1761)—quite independently of one another—**developed the first device for storing electricity**. The **Leyden jar stored a static electric charge** between two

electrodes on the inside and outside of a glass jar. It was not a battery, because it did not produce a charge itself—instead it stored a static charge built up by friction generators. However, it was a compact way of storing electricity and provided a useful and ready source of charge.



#### Leyden jar

The Leyden jar provided a way to store and build up a charge of static electricity, ready to be released.



#### Comets

Great comets—comets that are exceptionally bright—are seen less frequently than other comets. There are trillions of comets that remain undiscovered to this day.

1743 German astronomers Jan de Munck and Dirk Klinkenberg discover the **Great Comet**

1744 French mathematician César-François Cassini de Thury begins mapping France

1744 Swiss mathematician Leonhard Euler publishes important papers on **optics**

1745 Swiss naturalist Charles Bonnet publishes a **treatise on insectology**

1745 The **Leyden jar** is developed by two separate inventors: Ewald Georg von Kleist and Pieter van Musschenbroek

1745 French mathematician César-François Cassini de Thury creates the **Cassini map projection**

1745 French philosopher **Pierre Louis Maupertuis** proposes that **all life is descended from a common ancestor**

1746 English pharmacist **William Cookworthy** discovers **kaolin** in Cornwall, England